

Homework 3

ORF 245 - Fundamentals of Statistics

Due: Feb. 21st, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
 - Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).
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Exercise: 1. (Discrete Random Variables, PMF and CDF) Consider rolling a fair dice twice (independently), and let D be the random variable that equals the first roll minus the second roll. So, if the two dice come up $(1, 6)$ (i.e. the first die roll equals 1, and the second die roll equals 6) then $D = -5$; if the two dice come up $(6, 1)$ then $D = 5$.

- (a) Find the probability mass function (pmf) of D .
 - (b) Find and sketch the cumulative distribution function (cdf) of D .
 - (c) Compute the expected value of D , $\mathbb{E}[D]$ and the expected value of the *absolute value* of D , $\mathbb{E}[|D|]$.
 - (d) Compute $\text{Var}(D)$
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Exercise: 2. (Poisson Distribution) Let X have a Poisson distribution with parameter $\lambda > 0$. That is, the pmf of X is given by

$$p_X(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \text{for } k = 0, 1, \dots$$

For each part below, it will be helpful to use the series expansion for $e^\lambda = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$.

- (a) Verify that this is a valid pmf (i.e., that it's non-negative and sums to 1).
- (b) Show that mean of X is $\mathbb{E}[X] = \lambda$.
- (c) Show that variance of X is $\text{Var}(X) = \lambda$.
(Hint: One way to approach this is to first compute $\mathbb{E}[X(X-1)]$ – why is this easier? How does it help?)

Exercise: 3. (Validity of PDFs) Let

$$f(x) = \begin{cases} \frac{c}{x^2} & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

for some unknown constant c .

- (a) Find the value of c for which $f(\cdot)$ is a valid probability density function. Sketch or plot $f(\cdot)$.
- (b) Let X be a continuous random variable with the pdf above. Find the corresponding cdf $F_X(x)$. Sketch or plot $F_X(\cdot)$.
- (c) Compute $\mathbb{P}(1 \leq X \leq 2)$.

Exercise: 4. (A New Distribution?) Let X be a random variable with pdf

$$f(x) = \frac{1}{\sqrt{\pi}} \exp(-(x + \pi)^2)$$

- (a) Find $\mathbb{E}[X]$ and $\text{Var}(X)$.
- (b) Find $\mathbb{E}[X^2]$.
- (c) Let $Y = 2 - X/\pi$. Find $\mathbb{E}[Y]$ and $\text{Var}(Y)$.
- (d) Find $\mathbb{P}(X > \pi)$ in terms of $\Phi(z)$, the cdf of a standard normal random variable.

Exercise: 5. [OPTIONAL] (Conditional Geometric Distribution)

Note: This is an *optional* exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Recall that the probability mass function of a geometric random variable (parameterized by α) is

$$p_X(x; \alpha) = \begin{cases} (1 - \alpha)^{x-1} \alpha & \text{for } x = 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

Suppose $X \sim \text{Geo}(\alpha)$, i.e., the pmf of X is as given above, for some $\alpha \in (0, 1]$.

- (a) Find and sketch the cdf of X . Make sure to label your axes.
- (b) For a fixed $t \geq 1$, find $\mathbb{P}(X \geq t)$.
- (c) For a fixed $t \geq 1$, compute $\mathbb{P}(X = x | X \geq t)$ for $x \geq t$. I.e. compute the probability that X equals x conditioned on the event $X \geq t$.

This is called the *conditional pmf of X given $X \geq t$* , and is denoted $p_{X|X \geq t}(x; \alpha)$.

- (d) For some event A and discrete random variable X , define the *conditional expectation of X given A* as $\mathbb{E}[X|A] = \sum_{x \in \text{range}(X)} x \cdot p_{X|A}(x; \alpha)$, where $p_{X|A}(x)$ is the conditional probability mass function of X given event A .

Compute $\mathbb{E}[X|X \geq t]$.

(Hint: you can use the fact that $\mathbb{E}[X] = \frac{1}{\alpha}$.)

- (e) Provide a one-sentence interpretation of the results in part (d) – why might the geometric distribution be called “memoryless”?

(Hint: recall that geometric random variables are often used to model a “waiting time”.)